Lecture 9

Microscope Imaging (2)

Contents

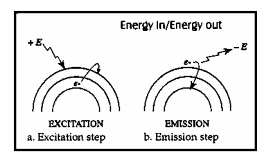
- Fluorescence Microscopy
  - Epi-fluorescence
  - CSLM
- Examples of microscopic imaging
- Visualization and staining
- 3D Visualization
- Image restoration
- MR microscopy

Fluorescence

Using the specimen as a light source

Principles of Fluorescence

- Basic principle: Energy in – Energy Out
- Fluorophore (molecule) gets exited by a high-energy source (arc-lamp, laser)

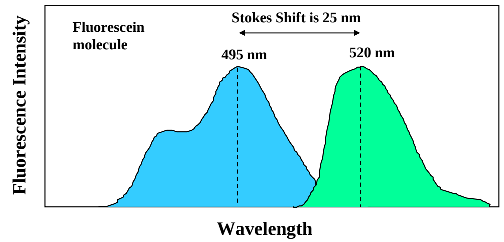


- Fluorescent dye: fluorochrome
- The **longer** the wavelength the **lower** the energy
- The **shorter** the wavelength the **higher** the energy
  - Note: UV light from sun causes the sunburn not the red visible light

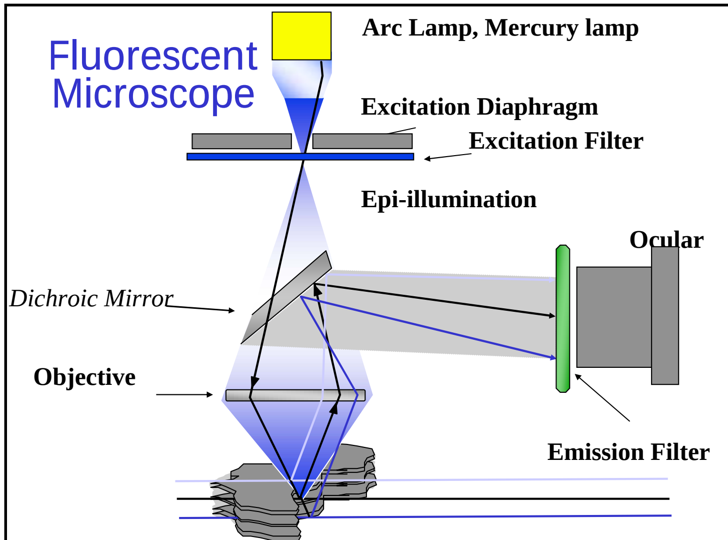
Fluorescence

Stokes Shift:

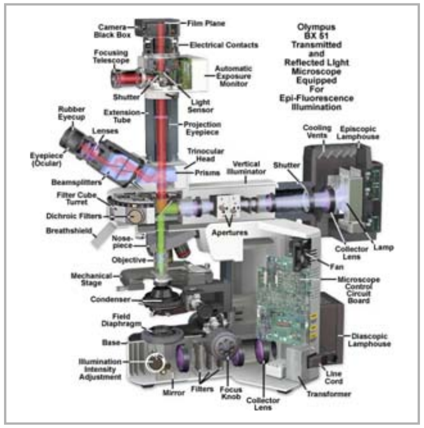
– the energy difference between the lowest energy peak of absorbance and the highest energy of emission

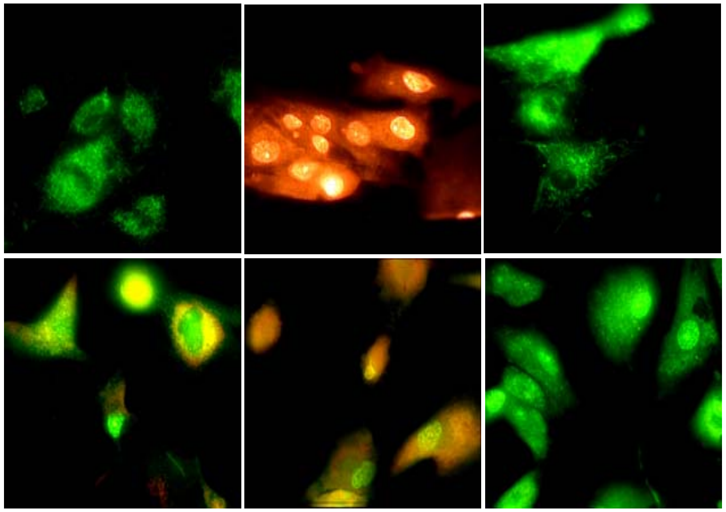


Fluorescent Microscope



Epi-fluorescent illumination

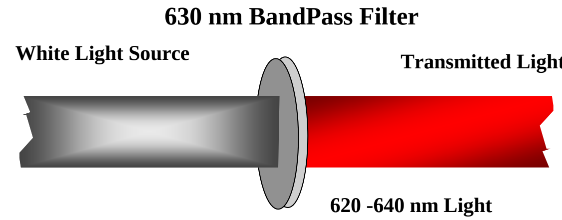




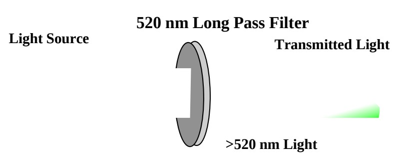
Images with fluorescent microscope

- High quality images
- Relative long range (working distance)
- Blurring
- Very specific contrast
- Suitable for analysis
- Drawbacks of fluorochromes

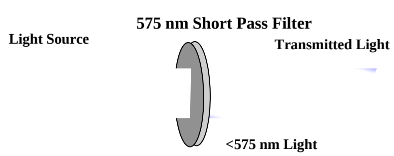
Standard Band Pass Filters



Standard Long Pass Filters



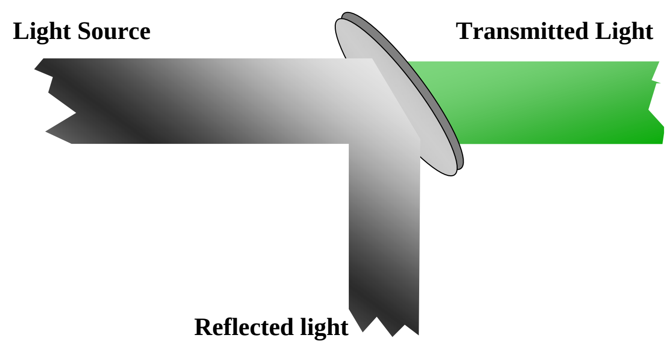
Standard Short Pass Filters



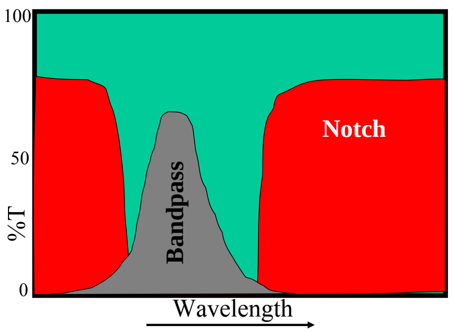
Optical Filters

510 LP dichroic Mirror

Dichroic Filter/Mirror at 45 deg

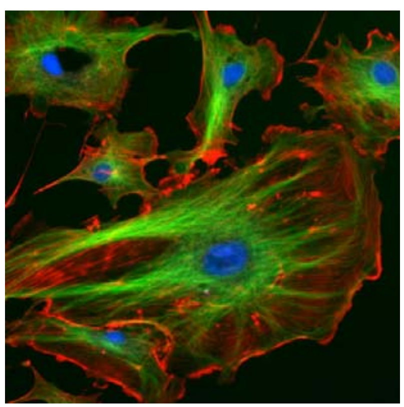


Filter Properties: Light Transmission



Filter Blocks for Fluorescence

This is not a standard image ...



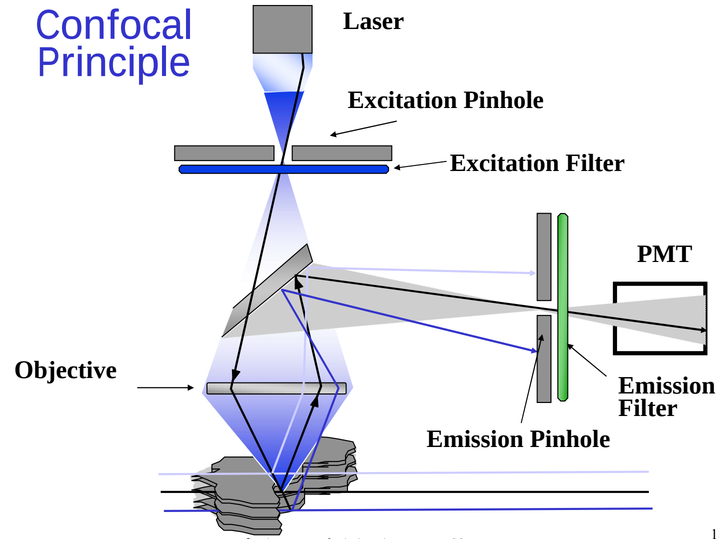
Proteins Probes

Probe	*Excitation*	*Emission*
FITC	488	525
PE	488	575
APC	630	650
PerCP™	488	680
Cascade Blue	360	450
Coumerin-phalloidin	350	450
Texas Red™	610	630
Tetramethylrhodamine-amines	550	575
CY3 (indotrimethinecyanines)	540	575
CY5 (indopentamethinecyanines)	640	670

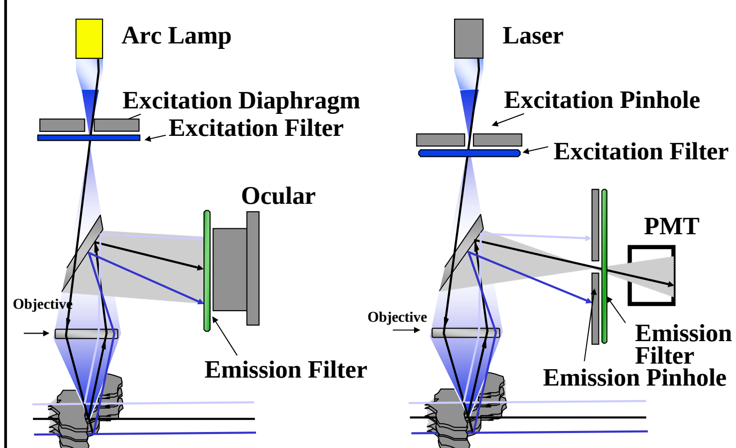
Confocal Laser Scanning Microscope

CLSM, uses fluorescence

Confocal Principle



Fluorescent Microscope | Confocal Microscope



Laser Light Path

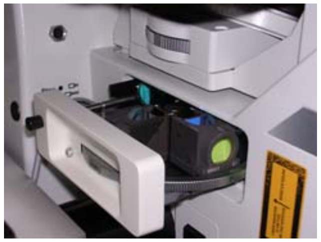
CLSM Light Source - Lasers

Laser	Abbreviation	**Excitation Lines**
• Argon	Ar	353-361, 488, 514 nm
• Violet Diode		405 nm
• Krypton-Ar	Kr-Ar	488, 568, 647 nm
• Helium-Neon	He-Ne	543 nm, 633 nm
• He-Cadmium	He-Cd	325 - 441 nm

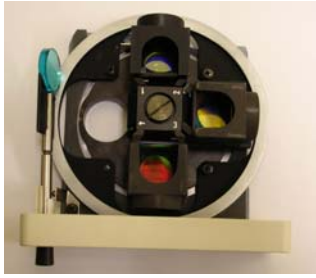
The CLSM

- Not just a microscope
- Microscope software
- Acquisition software
- All computer controlled

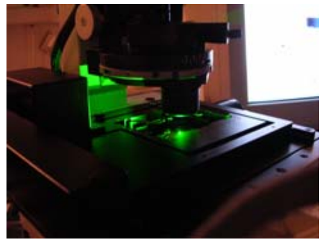
The CLSM, filter block



Inverted CLSM, filter block layout



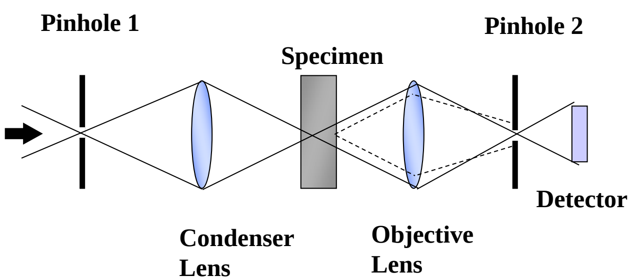
Inverted CLSM, components



Benefits of Confocal Microscopy

- Reduced blurring of the image from light scattering
- Increased effective resolution
- Improved signal to noise ratio
- Magnification can be adjusted electronically
- Clear examination of thick specimens
- Z-axis scanning
- Depth perception in Z-sampled images (optical sections)
- True 3D images, non-invasive imaging
- Suitable for image analysis (2D & 3D)

Image formation in CLSM

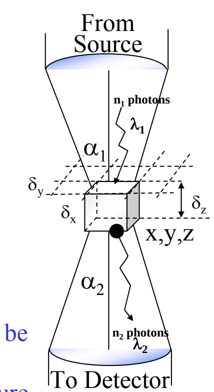


Modified from: Handbook of Biological Confocal Microscopy. J.B.Pawley, Plennum Press, 1989

Fundamental Limitations of Confocal Microscopy

PIXEL

VOXEL



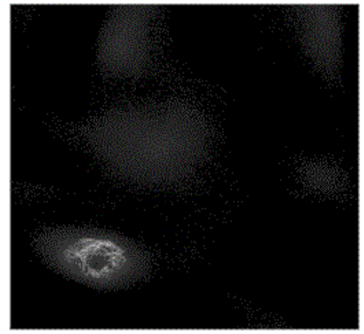
These limitations will be further discussed in image restoration lecture

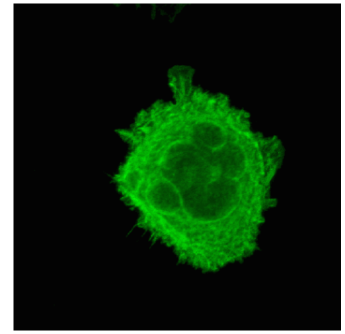
From: Handbook of Biological Confocal Microscopy. J.B.Pawley, Plennum Press, 1989

Specific contrast in CSLM

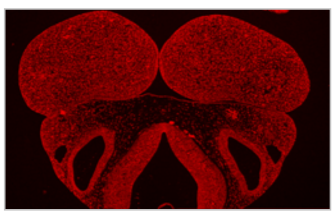
- Specific binding fluorchromes
  - Amino-acids
  - Lipids (membranes)
  - Nucleotides
- Fluorescent Immuno-histochemistry
  - Anti-bodies
- Fluorescent *in situ* Hybridization
  - Probes

Visualizing cells



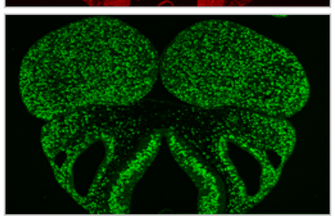


Channels in CLSM



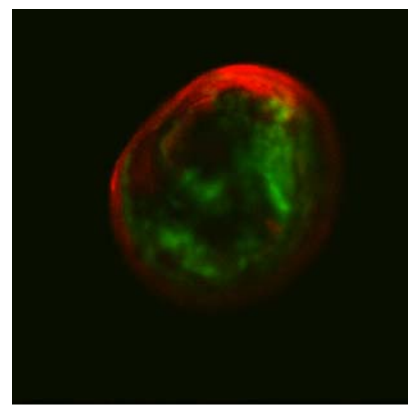
Bopro

Mouse embryo jaw



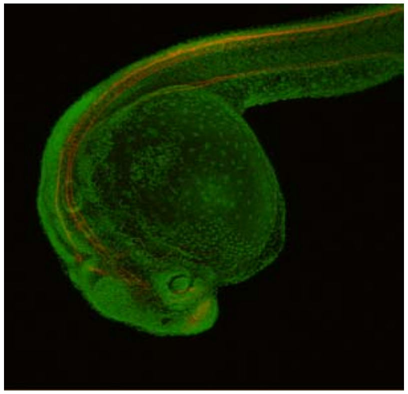
BrdU

Visualizing small specimens



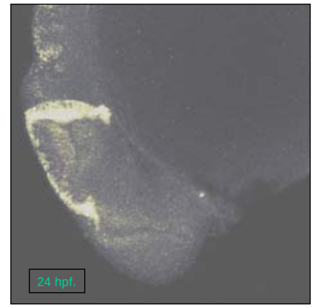
Mussel embryo

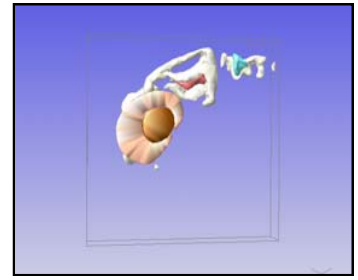
Visualizing larger specimens



Zebrafish 24 hrs. pf.

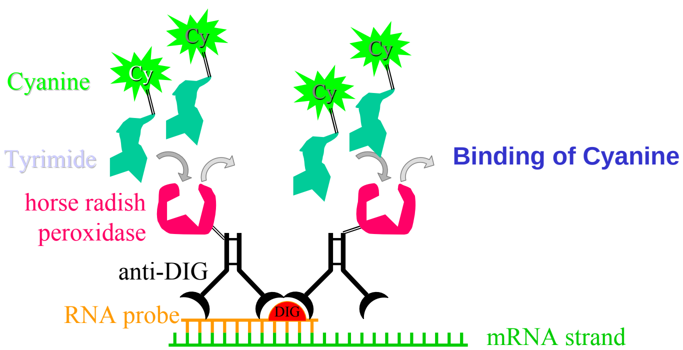
FISH in CLSM (weak signal)

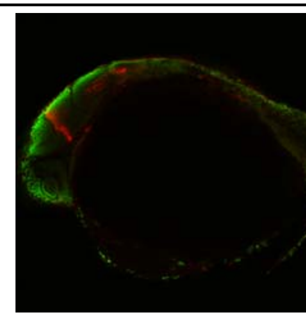


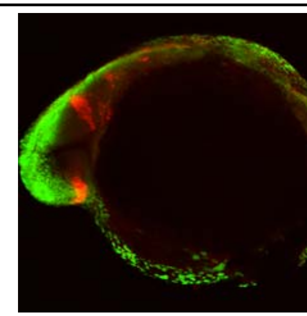


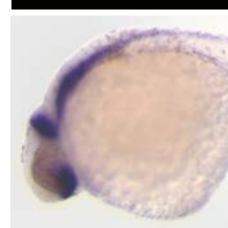
Zebrafish 24 hrs. pf.

FISH + Tyramide Signal Amplification









Pax2.1

Zebrafish 24 hrs pf.

Welten, et al. 2006

GFP Probes

GFP = Green Fluorescent Protein (family of probes)

- GFP is extracted from chemiluminescent jellyfish *Aequorea victoria*
- Excitation:
  - maxima at 395 and 470 nm
  - quantum efficiency is 0.8
  - peak emission at 509 nm
- Application: reporter gene for assay of promoter activity
- requires no added substrates

CLSM Images: *NeuroGenine Expression*

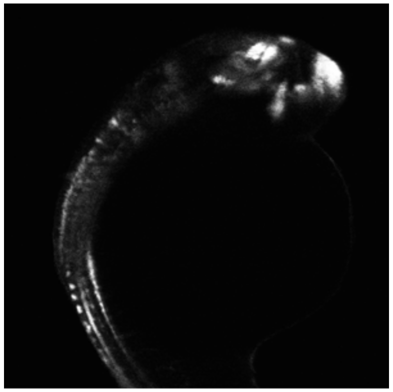
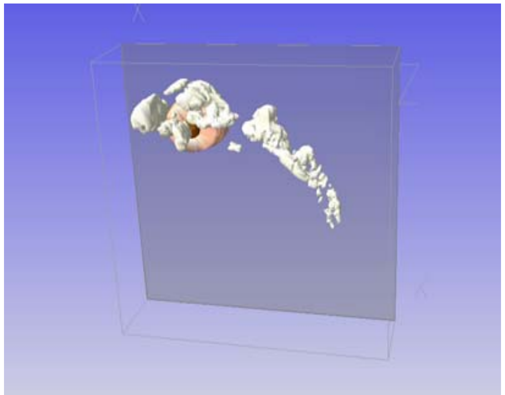


Image Reconstruction

- Reconstruction is the process of attempting to recreate the original signal given a corrupted one
- Terms:
  - Scene: the "real world"
  - Image: a (possibly corrupted) representation of a scene
- Image reconstruction attempts to recreate the scene from an image

Reconstruction : *Neurogenine expression*



Knowledge Reconstruction

- Reconstruction algorithms usually use one or more of
  - Knowledge about the process that corrupted the image
  - Knowledge about properties of the original scene
- Examples:
  - Deconvolution requires knowledge of the point spread function
  - Weiner filtering requires an estimate of the strength of the noise

Knowledge about the corruption process puts limits on reconstruction

Deconvolution

- Deconvolution is the reverse operation of convolution.
- Deconvolution a.k.a. Deblurring
- Removing "out of focus" contributions to image formation
- The lens and its properties need be modeled:
  - Point spread function (PSF)
  - Estimation of PSF

Point Spread Function

Conventional Microscopy

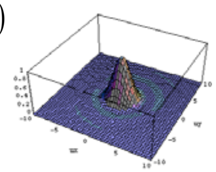
- Systems characterization Rayleigh Distance
- Optical system: point spread function: *psf* (x,y)
- *psf* (x,y) intensity distribution of imaging small point-like object. This looks like:

$$psf(x,y) = \left(\frac{2 \cdot J_1(v)}{v}\right)^2$$

- where v:

$$v = \frac{2 \cdot \pi}{\lambda} \cdot r \cdot \sin(\beta)$$

J_1 first order Bessel function
r circle radius of object at x,y
λ wavelength of light



psf and Modulation Transfer

- The Fourier transform of the *psf* (x,y) is referred to as the Modulation Transfer Function (MTF):

$$MTF = \Im[psf(x,y)]$$

Bandwidth of MTF relates to Rayleigh criterion

- Frequency distribution in MTF helps to understand optical system
- In CLSM laser beam and emission light is focused in diffraction spot, hence the psf of a confocal system is the square of a "incoherent light" system:

$$psf_{confocal}(r) = \left(\frac{2 \cdot J_1(v)}{v}\right)^4$$

Computing Deconvolution

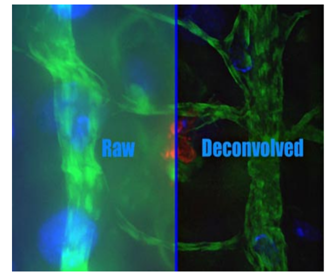
- Compute PSF
  - Quantitative compensation for deblurring
  - PSF describes light passing the system's optics
  - Object+PSF = out of focus haze
  - Iterative algorithm, Maximum Likelihood estimation

$$\hat{\lambda}^{(k+1)}(x) = \hat{\lambda}^{(k)} \cdot \int_{R^3} \frac{h(u-x)}{\int_{R^3} h(u-z)\hat{\lambda}^{(k)}(z)dz} \mu_n(u)du$$

  - μ (x): collected data,
  - λ (x): most likely caused observed data
  - *h* (x): the Point Spread Function, estimated from point source

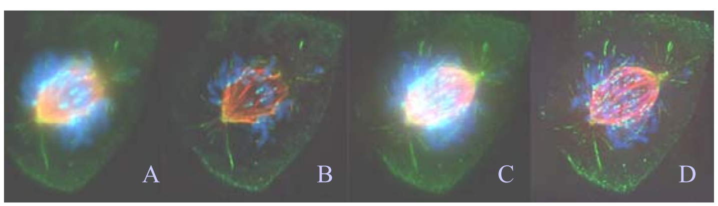
2D Deconvolution

- Fluorescence Imaging:



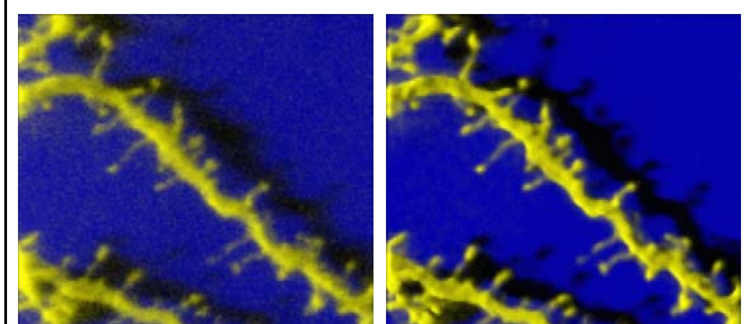
- Lots of dedicated software

3D Deconvolution



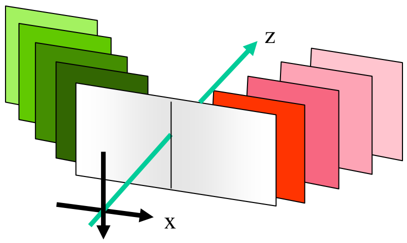
- Mitotic spindle, before (A,C) and after deconvolution (B,D)
- This is one optical section (A,C), 3 channels!
- Deconvolution is applied on a 3D image!
  - computationally intensive
  - requires input of adjacent layers (participates in blur)

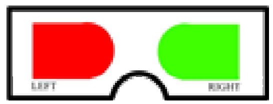
Example with dendrite

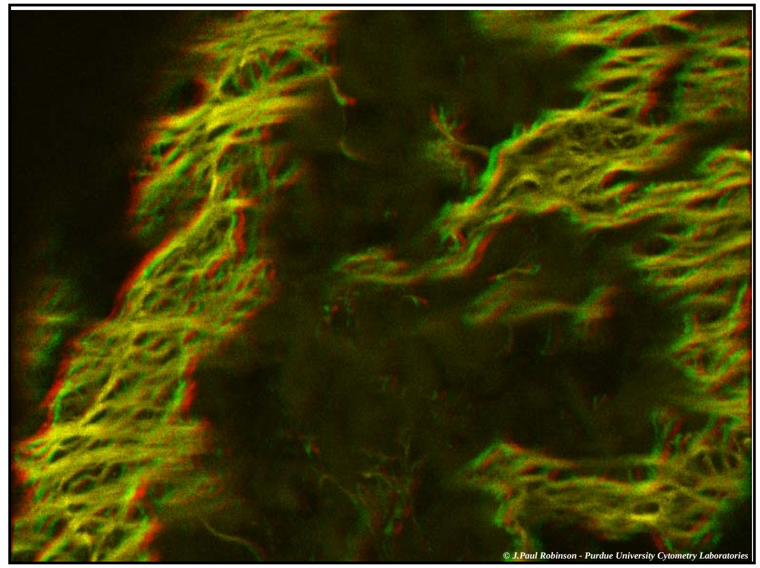


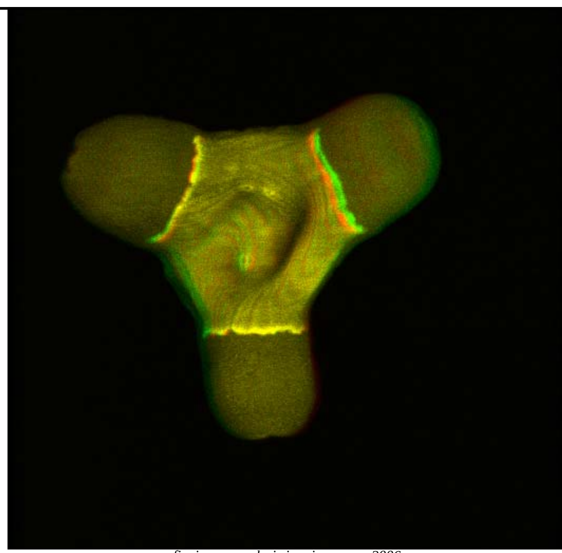
Stereo Images

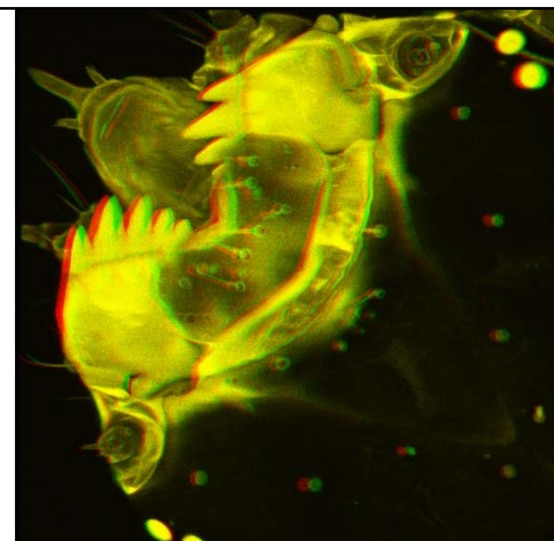
- methods for creating stereo pairs:
  - different colors for each image: red-green which produce a monochrome 3D image
  - side-by-side display of both images

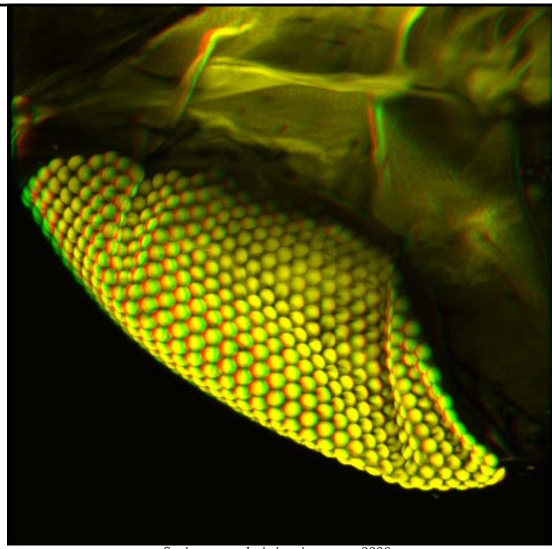


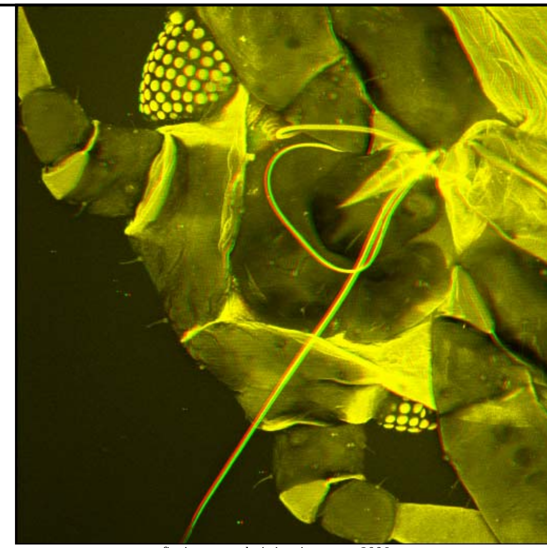








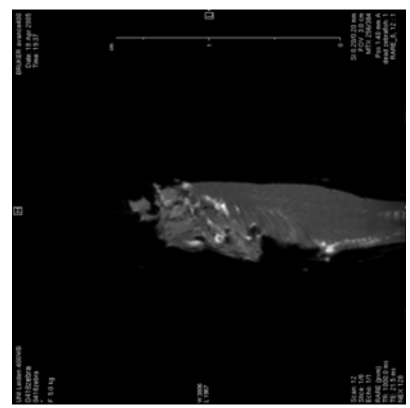




Vivo 3D Imaging

- Micro MRI or μMRI or MR microscopy
  - Imaging using Magnetic Resonance
  - Relaxation times (T1,T2) for contrast
    - physical features
  - Non-Invasive – living animals
  - μMRI is similar to clinical MRI
    - optimized for small structures (≤ 100 mm)
    - higher magnetic field
  - Small animal imaging
  - Special flow chamber & Coil

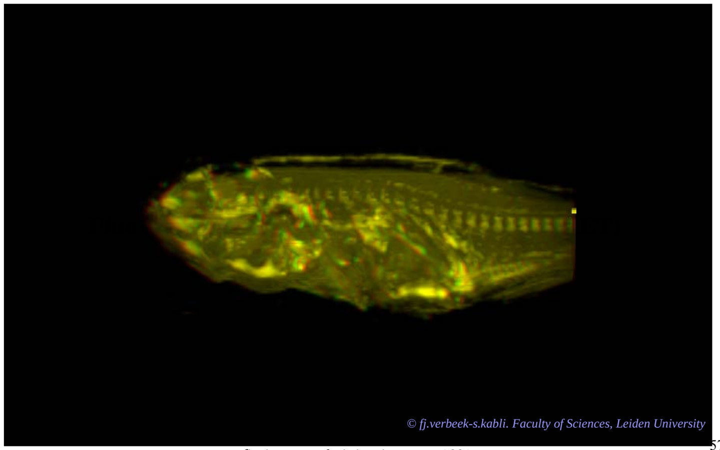
MR-Microscopy images



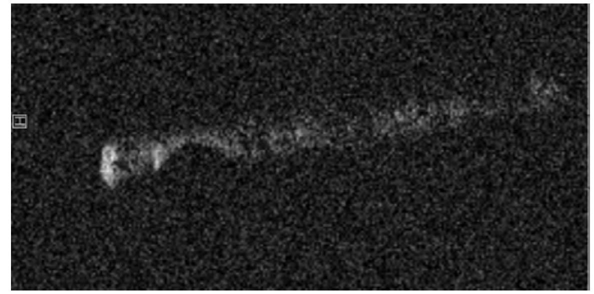
9.4T field

Kabli, Alia, Verbeek

Simple 3D visualization MR-M

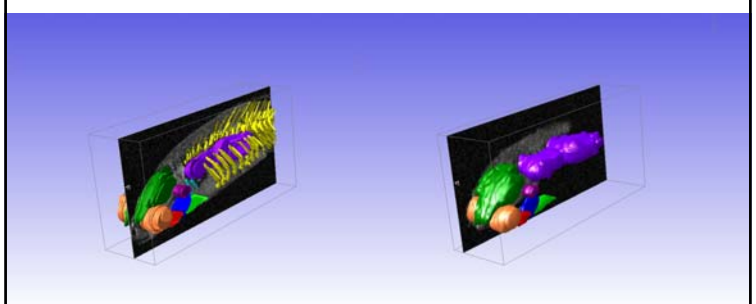


MR images - stack



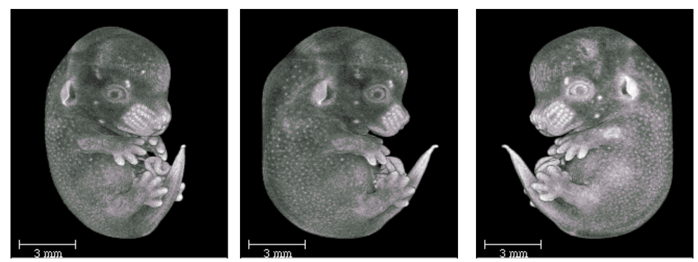
Kabli, Alia, Verbeek

3D Modeling from µMRI



Kabli, Alia, Verbeek, 2006

3D Visualization MR Microscopy

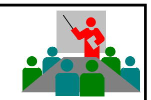


Volume rendering: iso-surfaces

Microscopy Not Discussed

- Dark Field Microscopy
- Phase Contrast
- Difference Interference Contrast (DIC)
- Nomarski Microscopy
- Near field Microscopy
- Multi Photon Microscopy
- Atomic Force Microscope

Review



- Microscopes
  - Fluorescence
  - CLSM
  - Building blocks
- Specific contrast agents for fluorescence
- Simple visualization techniques
- Image restoration by deconvolution
- MR micrscopy

Acknowledgements

- Museum of microscopy.
- Some slides adapted from:
  - Dr. J. P. Robinson, Purdue University
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  - G. Lamers, IBL
  - FJV Research material
  - Stereo Images JP Robinson